EVERY ALGEBRAIC GROUP SCHEME OVER A FIELD IS QUASI-PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. No linearity or affineness assumption on the algebraic group is introduced; the target is quasi-projectivity for arbitrary algebraic group schemes over the given field.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Algebraic group schemes, connected components, Cartier smoothness in characteristic zero and smoothness of reduced groups over a perfect field, descent of ampleness, regular-local factoriality, etale charts, divisor norms and rational equivalence of Picard classes over an open affine-space base.

Source reference. Original lemma 0BDC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14310–14310. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 0B7R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14114–14114. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 047N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5514–5514. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 0BD4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14302–14302. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0B42: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13985–13985. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 047R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5518–5518. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:groupoids.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 047P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5516–5516. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 056S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6720–6720. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AG0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13178–13178. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BCW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14294–14294. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 047L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5512–5512. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 054L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6644–6644. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0BCX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14295–14295. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AGA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13188–13188. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BEH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14350–14350. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 01WX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2498–2498. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains

the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 01PS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2241–2241. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let k be a field. Let G be an algebraic group scheme over k. Then G is quasi-projective over k.*

Proof. By Varieties, Lemma 0BDC we may assume that k is algebraically closed. Let $G^0 \subset G$ be the connected component of G as in Proposition 0B7R. Then every other connected component of G has a k-rational point and hence is isomorphic to G^0 as a scheme. Since G is quasi-compact and Noetherian, there are finitely many of these connected components. Thus we reduce to the case discussed in the next paragraph.

Let G be a connected algebraic group scheme over an algebraically closed field k. If the characteristic of k is zero, then G is smooth over k by Lemma 047N. If the characteristic of k is $p > 0$, then we let $H = G_{red}$ be the reduction of G. By Divisors, Proposition 0BD4 it suffices to show that H has an ample invertible sheaf. (For an algebraic scheme over k having an ample invertible sheaf is equivalent to being quasi-projective over k, see for example the very general More on Morphisms, Lemma 0B42.) By Lemma 047R we see that H is a group scheme over k. By Lemma 047P we see that H is smooth over k. This reduces us to the situation discussed in the next paragraph.

Let G be a quasi-compact irreducible smooth group scheme over an algebraically closed field k. Observe that the local rings of G are regular and hence UFDs (Varieties, Lemma 056S and More on Algebra, Lemma 0AG0). The complement of a nonempty affine open of G is the support of an effective Cartier divisor D. This follows from Divisors, Lemma 0BCW. (Observe that G is separated by Lemma 047L.) We conclude there exists an effective Cartier divisor $D \subset G$ such that $G \setminus D$ is affine. We will use below that for any $n \geq 1$ and $g_1, \ldots, g_n \in G(k)$ the complement $G \setminus \bigcup Dg_i$ is affine. Namely, it is the intersection of the affine opens $G \setminus Dg_i \cong G \setminus D$ in the separated scheme G.

We may choose the top row of the diagram

$$\begin{array}{ccccc} G & \xleftarrow{j} & U & \xrightarrow{\pi} & \mathbf{A}^d_k \\ & & \uparrow & & \uparrow \\ & & W & \xrightarrow{\pi'} & V \end{array}$$

such that $U \neq \emptyset$, $j : U \to G$ is an open immersion, and π is étale, see Morphisms, Lemma 054L. There is a nonempty affine open $V \subset \mathbf{A}^d_k$ such that with $W = \pi^{-1}(V)$ the morphism $\pi' = \pi|_W : W \to V$ is finite étale. In particular π' is finite locally free, say of degree n. Consider the effective Cartier divisor

$$\mathcal{D} = \{(g, w) \mid m(g, j(w)) \in D\} \subset G \times W$$

(This is the restriction to $G \times W$ of the pullback of $D \subset G$ under the flat morphism $m : G \times G \to G$.) Consider the closed subset[1] $T = (1 \times \pi')(\mathcal{D}) \subset G \times V$. Since

[1]Using the material in Divisors, Section 0BCX we could take as effective Cartier divisor E the norm of the effective Cartier divisor $\mathcal{D}$ along the finite locally free morphism $1 \times \pi'$ bypassing some of the arguments.

π' is finite locally free, every irreducible component of T has codimension 1 in $G \times V$. Since $G \times V$ is smooth over k we conclude these components are effective Cartier divisors (Divisors, Lemma 0AGA and lemmas cited above) and hence T is the support of an effective Cartier divisor E in $G \times V$. If $v \in V(k)$, then $(\pi')^{-1}(v) = \{w_1, \ldots, w_n\} \subset W(k)$ and we see that

$$E_v = \bigcup\nolimits_{i=1,\ldots,n} Dj(w_i)^{-1}$$

in G set theoretically. In particular we see that $G \setminus E_v$ is affine open (see above). Moreover, if $g \in G(k)$, then there exists a $v \in V$ such that $g \notin E_v$. Namely, the set W' of $w \in W$ such that $g \notin Dj(w)^{-1}$ is nonempty open and it suffices to pick v such that the fibre of $W' \to V$ over v has n elements.

Consider the invertible sheaf $\mathcal{M} = \mathcal{O}_{G \times V}(E)$ on $G \times V$. By Varieties, Lemma 0BEH the isomorphism class $\mathcal{L}$ of the restriction $\mathcal{M}_v = \mathcal{O}_G(E_v)$ is independent of $v \in V(k)$. On the other hand, for every $g \in G(k)$ we can find a v such that $g \notin E_v$ and such that $G \setminus E_v$ is affine. Thus the canonical section (Divisors, Definition 01WX) of $\mathcal{O}_G(E_v)$ corresponds to a section s_v of $\mathcal{L}$ which does not vanish at g and such that G_{s_v} is affine. This means that $\mathcal{L}$ is ample by definition (Properties, Definition 01PS). □

References